

MyDesign Portfolio

Element D Initial Template

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

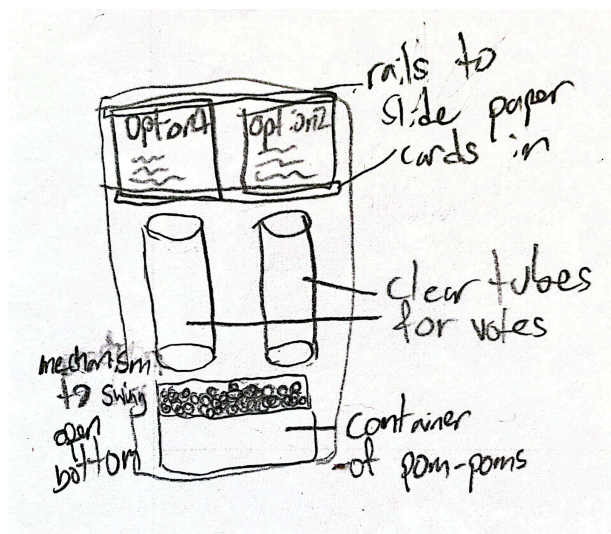
This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

John's First Design

Date: March 20

Description: A main container of pom-poms is the base of the device. A wooden backboard is attached to the container. Above the container attached to the backboard are two, hollow acrylic/plastic cylinders. Above the cylinders are rails where you can slide cards inside to easily switch between questions. The kids would take a pom pom from the main container and drop it in one of the tubes to vote. At the bottom of each tube, you can swing the bottom open to release all the pom poms into the main container when a vote is over to reset for the next question.

Drawing:



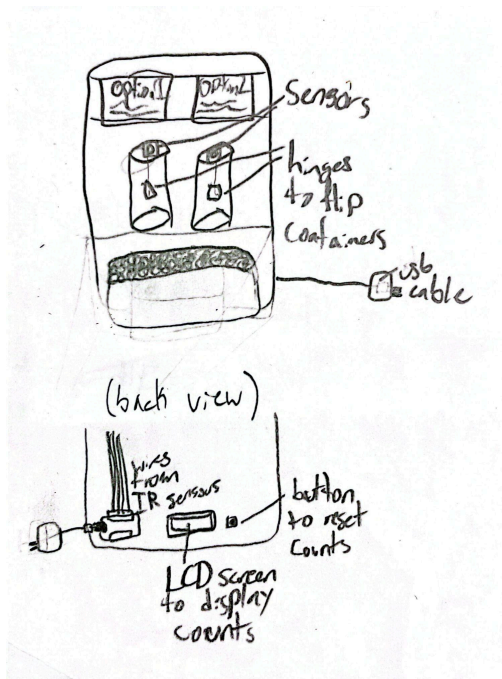
John's Second Design

Date: March 22

Description: The design is exactly the same as my first design, except for two changes to the acrylic cylindrical containers.

- Instead of having a mechanism to swing the bottom open, the containers can be attached to a hinge and can be spun upside down to reset the votes. It would have a mechanism to lock into place at the default upright position to prevent it from accidentally flipping.
- Each container will have an IR sensor or other object detection sensor at the top lip, which will get triggered every time a pom-pom is dropped inside. The IR sensors are connected to an arduino or other microcontroller which is housed somewhere in the back of the device. The microcontroller will keep two counts of how many times a pom-pom was dropped in each container. Also at the back of the device would be an LCD screen to display the counts and a button to reset both counts. It is powered by an outlet through a usb power adapter. This means that at the end of the vote, it can have exact data rather than having to eyeball how many pom-poms are in each container.

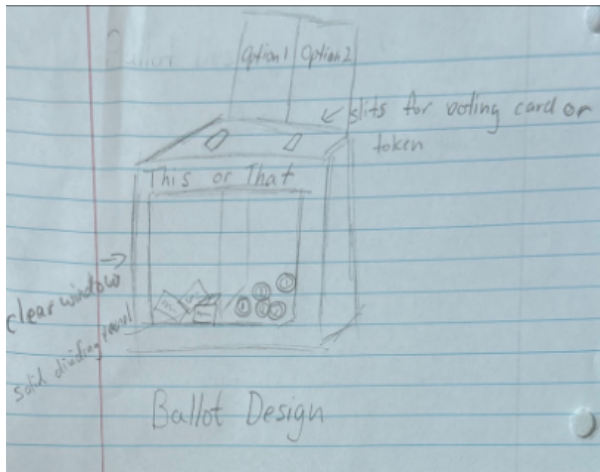
Drawing:



Sofia's Design 2 (the one we chose)

Date: March 21

Description: A wooden box with two slits on the top and the *THIS* or *THAT* on top. Can use little tokens or paper cards to *vote*. The front of the box can be clear so kids can see which is more popular. To reset the vote cards/token, there will be a open side to remove and replace them.



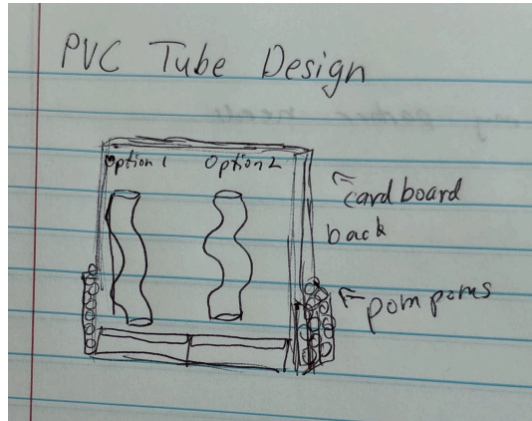
Sketch:

Sofia's first Design

Date: March 20 2026

Description: The design is built on a board of cardboard. At the top of the board, you would write (on sticky notes) the two different prompts. Underneath the options, there would be two clear pvc tubes that drop into boxes for the respective choices. The kids would use one of the two different color pom poms to choose.

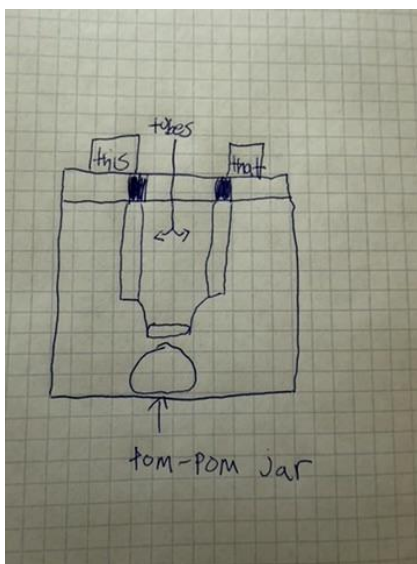
Drawing:



James's Design

Date: March 21, 2026 (sketch uploaded 4/14/26)

Description: This design starts with a standard cardboard box with two holes cut in it, and a tube leading through each hole into two different jars, leading to the THIS or THAT questions. Pom-poms will be in each tube with different THIS or THAT questions. It will be open-windowed.



Sketch:

Matthew's Design

Date: March 22 (sketch uploaded 4/14/26)

Description: Have a dry erase board attached to two buckets(Containers), pom poms are kept in a plastic cup, they are 2 colors red and blue. When voting is done just empty the buckets and count how many colored pom poms are there in each. The dry erase board is so you don't waste paper.



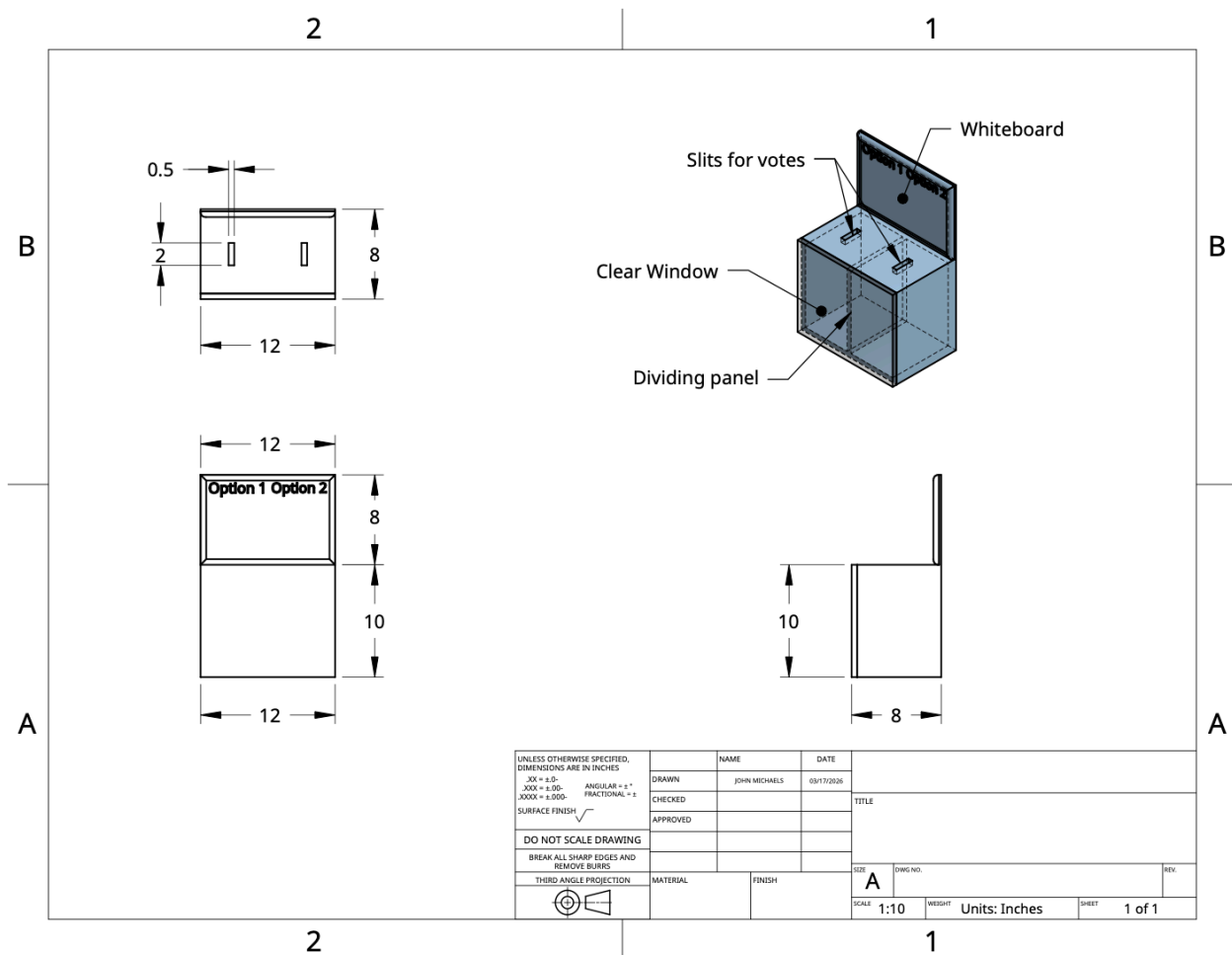
Design Idea Comparison and Selection (D1, D2)

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

	Weight	Sofia's First Design	Sofia's Second Design	John's First Design	John's Second Design	James' Design	Matthews' Design
Engagement/Interactivity	0.2	4	4	3.8125	3.5625	3.625	3.375
Cost/Construction	0.09	3	3.575	3.5	2.825	3	3.5
Durability	0.18	4	3.875	3.375	3.5	3.0625	3.75
Ease of Resetting/Use	0.21	3.8125	3.8125	3.875	3.625	3.375	3.625
Functional Performance	0.17	4	4	3.5	3.875	3.875	3.5
Safety	0.15	4	4	3.875	3.75	3.875	3.9375
Score		22.8125	23.2625	21.9375	21.1375	20.8125	21.6875

Design Blueprint (D4)

Design Description: We chose this design since it met most of the requirements we created and the constraints that were given. The clear front panel allows the kids to see the votes which creates engagement. Being made mostly out of wood made it the most secure and durable so the kids would not hurt themselves. Having the options displayed at the top allow the kids to easily see what the choices are.



CAD link:

<https://cad.onshape.com/documents/d0a02716bd42f458a3630cfb/w/d846f33ec499afc9bb9537d8/e/801b7674192f0b53ad23feba?renderMode=0&uiState=69b9a9168cb7e5e662222915>

Manufacturing Methods:

1. Use woodglue to join the wood boards together.
2. Cut one wooden board with two 2inch slits in the middle.
3. Attach plexiglass to fit the front window and glue to frame.

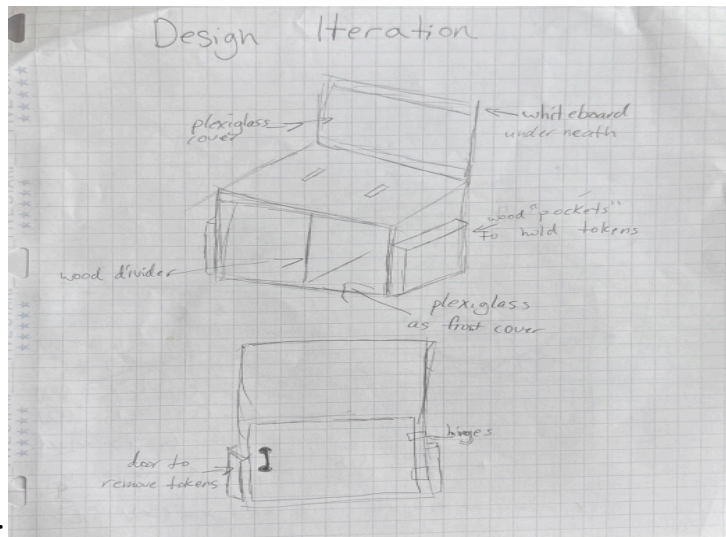
4. Attach a wooden board as a divider in the middle of the box.
5. Attach the top board with slits.
6. Glue 18x12in board to the back.
7. Cut a door in the large backboard and add the hinges to the right to hold the door.
8. Use the wood glue to attach the back board.
9. Hot glue the felt to create pockets on the sides of the box.
10. Take a clear folder sleeve, and cut and glue to the backboard, and slide the paper cards in between the sheet.

Supply List and Costs:

Material	Amount	Cost
Wood	2(8x12in) 2(8x10in) 2(10x12)	12.99
Plexiglass	1(8x12in)	9.98
Paper Cards	2	0.99
Wood Glue	1	7.99
Small handsaw (maybe)	1	We need this now, but it's free because everyone has it.
Marker	1	free(household item)
Hammer	1	Free (household item)
Wooden Tokens	100(1.5in)	13.99
Felt	2(7x3in)pieces	3.99
Hot glue	2-3 Sticks	2.99
Total Cost	53.38	

Here is a link to the Gantt Chart: [+ Element D Gantt Chart Share](#)

Second Iteration



New Sketch:

Description: Similar to the original design, but with pockets on the side to hold the tokens before voting. The back will have a door cut out on the back for easy access to remove the tokens attached with hinges. It will allow the librarian to pull out the wooden tokens and keep using them for future voting. The pockets on the sides will be made of felt for ease of use and not to add too much weight to the design. The top board will have a clear folder cover glued to the back board on the top to protect the paper card writing. You will be able to slide the cards in and out and easily write on them.

Here is the link to our revised CAD drawing.

<https://cad.onshape.com/documents/d0a02716bd42f458a3630cfb/w/d846f33ec499afc9bb9537d8/e/d6af6f76170b9ba6d8f37ad0>